

EduGuardian AI — Academic Risk Intelligence

Student Academic Risk Calculation Framework & Multi-Agent Early Detection Architecture

1. Executive Overview

The EduGuardian Risk Intelligence Engine continuously analyzes real-time behavioral signals across four core academic dimensions to proactively detect student academic distress before failure occurs. Students are categorized into **High Risk**, **Medium Risk**, and **Low Risk** tiers, providing live visibility for class admins while automatically steering adaptive AI coaching.

2. Four Core Signal Dimensions & Formulas

Signal Dimension	Data Source	Calculation Formula	Critical Threshold	Weight
1. Attendance Rate	attendance_records	(Classes Attended / Classes Held) * 100	< 65.0%	35%
2. Quiz / Exam Score	quiz_results	(Total Marks / Max Marks) * 100	< 50.0%	30%
3. Assignment Health	assignment_submissions	Count of Overdue / Missed Submissions	>= 2 Missed	20%
4. LMS Engagement	lms_activity	Session frequency & total active study minutes	Inactivity > 10 days	15%

3. Risk Classification Rules & Thresholds

Risk Tier	Trigger Criteria & Academic Indicators	Recovery Probability	System Action & Interventions
HIGH RISK	- Attendance < 65% OR drop > 15% - Quiz average < 50% - Missed assignments >= 2	< 50% (e.g., 38%-48%)	- Flagged on Class Admin Dashboard - Priority Mentor Assignment - Recovery Coach intervention triggered
MEDIUM RISK	- Attendance 65% - 79% - Quiz average 50% - 74% - Missed assignments = 1	50% - 80% (e.g., 65%-78%)	- Academic Watchlist monitoring - Study Planner catch-up roadmap - Automated remedial quiz suggestions
LOW RISK	- Attendance >= 80% - Quiz average >= 75% - Missed assignments = 0	> 80% (e.g., 85%-98%)	- Positive habit reinforcement - Advanced concept challenges - Standard progress tracking

4. Explainable Persistence Schema (PostgreSQL)

All risk computations are persisted in the `risk_predictions` table with human-readable support signals and SHAP feature importance:

```
CREATE TABLE risk_predictions (  
  student_id          INT REFERENCES students(id) ON DELETE CASCADE,  
  risk_level          VARCHAR(20) CHECK (risk_level IN ('low', 'medium', 'high')),  
  recovery_probability DECIMAL(5,2),    -- e.g. 42.50%  
  support_signal       TEXT,             -- Human-readable explainable trigger  
  attendance_change    DECIMAL(6,2),    -- Delta percentage trend (e.g. -18.00%)  
  lms_activity_change  DECIMAL(6,2),    -- Delta LMS usage frequency  
  missed_assignments  INT DEFAULT 0,    -- Count of overdue tasks  
  shap_explanation      JSONB            -- Feature weight vector explaining decision  
);
```

5. AI Multi-Agent Coaching Integration

- **Recovery Coach Agent (:8003):** Reads student risk metrics and crafts non-stigmatizing, encouraging check-in messages tailored to rebuild confidence.
- **Study Planner Agent (:8002):** Dynamically recalculates study roadmaps and milestones when missed assignments or declining quiz trends are detected.
- **Student Insight Agent (:8001):** Adjusts teaching depth and generates scaffolded practice questions for topics where student scored below 50%.